

09.1

Scientists investigated chloroplasts taken from leaf cells of the species *Nicotiana tabacum*.

The scientists took chloroplasts from leaf cells of a healthy plant and from leaf cells of a mutant plant.

Chloroplasts are contained in several different types of specialised cells in *N. tabacum* leaves.

What terms are used to describe an organisation of specialised cells in an organism when:

- all the cells are of one type
- the cells are of more than one type?

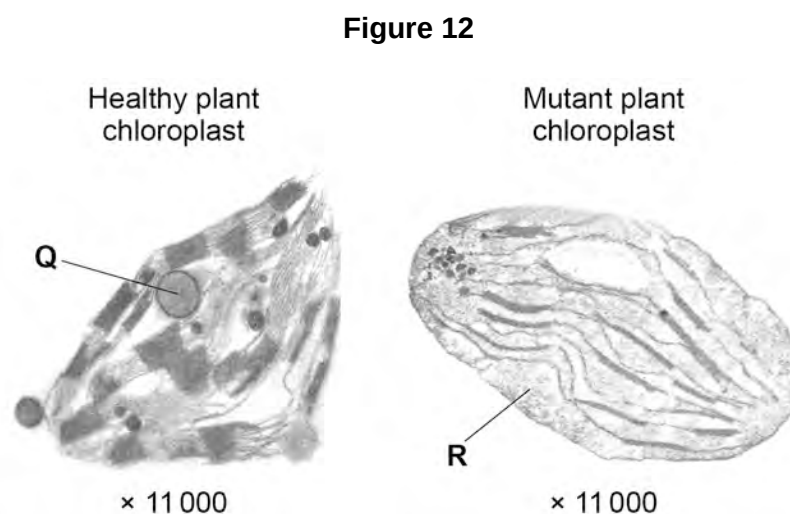
[1 mark]

Cells of one type _____

Cells of more than one type _____

09.2

Figure 12 shows images of chloroplasts from *N. tabacum* cells observed with an electron microscope.



Identify the parts labelled **Q** and **R**.

[1 mark]

Q _____

R _____

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Use the images in **Figure 12** to predict how photosynthesis could be affected in the mutant plants.

Justify your answer.

[2 marks]

Question 9 continues on the next page

09.4

Scientists investigated the rate of photosynthesis in healthy plants and in mutant plants.

Table 6 shows the scientists' results.

Chlorophyll a and chlorophyll b are two types of chlorophyll molecules used in photosynthesis.

Table 6

	Chlorophyll concentration / $\mu\text{g cm}^{-3}$	Chlorophyll a : chlorophyll b ratio	Rate of photosynthesis / arbitrary units
Healthy plant	45	2 : 1	1
Mutant plant	9	6 : 1	5

The scientists' results did not support what they predicted from **Figure 12** (on page 28) would be the effect of the mutation on photosynthesis.

Use information in **Table 6** to describe **and** explain the effect of the mutation.

[3 marks]

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